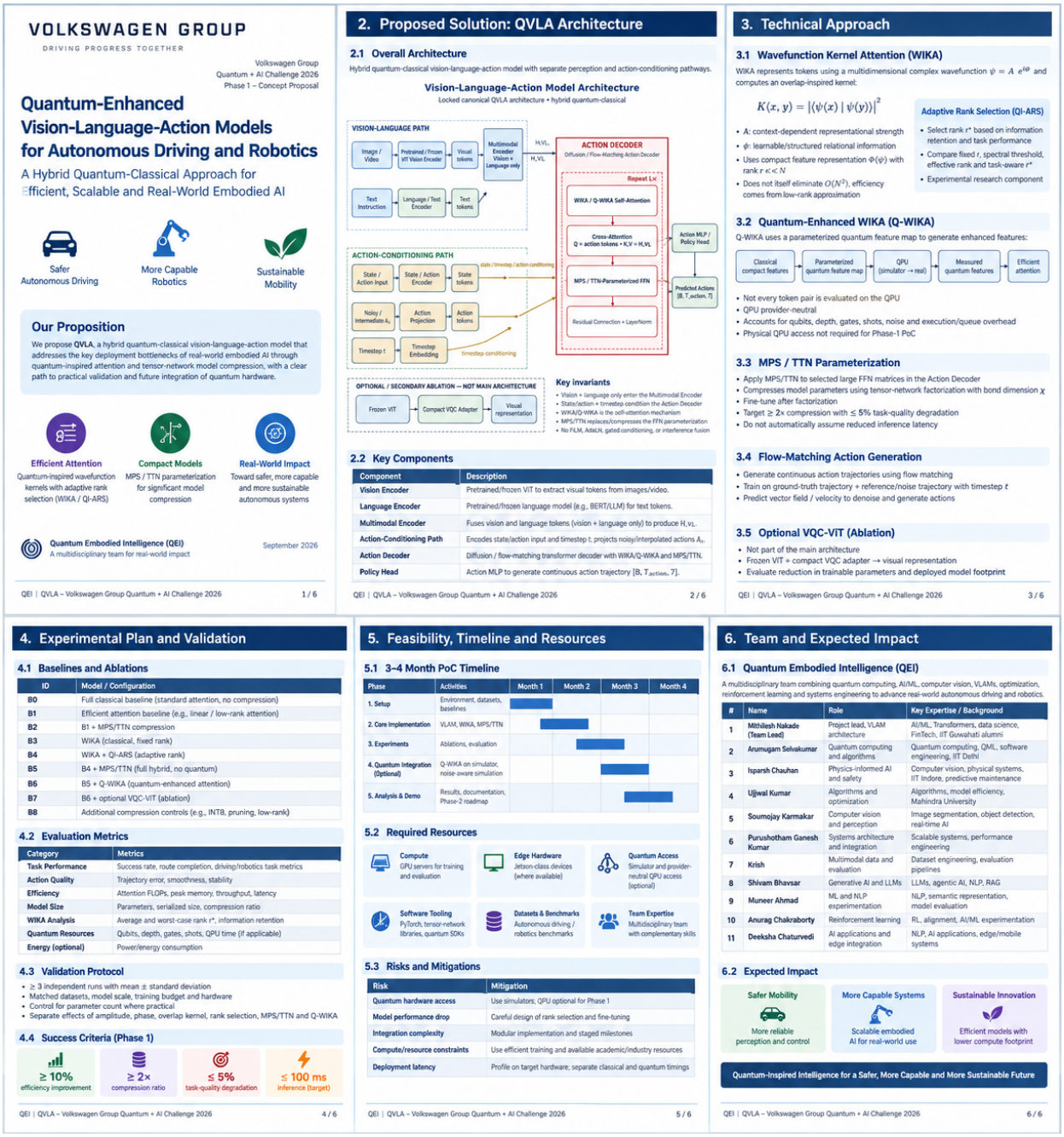
QVLA — Supplementary Technical Package

Technical visuals and classical-VLA reference context supporting the Phase 1 concept proposal

1. End-to-End Project Overview



Supplementary visual overview of the proposed QVLA system, its quantum-inspired components, and the intended autonomous-driving / robotics application context.

Classical VLA reference	QVLA extension
Flow-matching / diffusion-type VLA: noisy action trajectory + timestep → full/bidirectional self-attention → cross-attention → FFN → continuous action trajectory.	Replace the conventional self-attention with WIKA / Q-WIKA and selected dense FFN parameterization with MPS/TTN, while retaining the continuous Flow-Matching action pathway.
Autoregressive / sequential VLA: previous action → causal self-attention → cross-attention → FFN → next action, with sequential feedback.	QVLA targets the continuous-action flow-matching pattern rather than the autoregressive pathway; the distinction is retained as architectural context, not as an additional QVLA branch.

Source basis: the attached VLA architecture notes distinguish these two common encoder-decoder patterns and identify the flow-matching path as the continuous multi-action generation pattern used in the QVLA concept.

2. Canonical QVLA Architecture

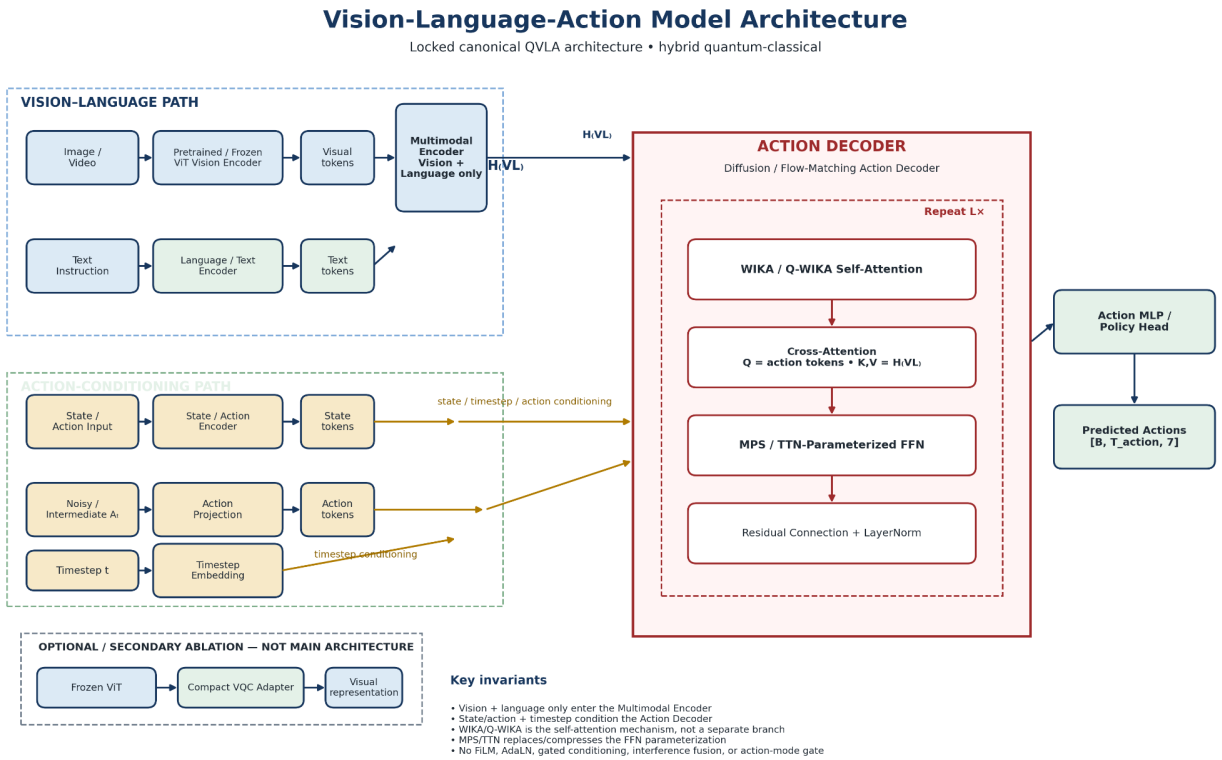


Figure S1. Locked canonical architecture used by the main Phase 1 proposal. The diagram is reproduced without architectural modification.

Classical VLA Flow Reference

Pattern	Action input	Self-attention	Output behavior
Flow-matching / diffusion-type	Noisy/intermediate action trajectory + timestep	Full bidirectional	Predict vector field; refine trajectory; multi-action generation
Autoregressive / sequential	Previous action	Causal masked	Predict next action; feed prediction back sequentially
QVLA target	Noisy/intermediate action trajectory + timestep	WIKA / Q-WIKA full-sequence mechanism	Flow-matching continuous action generation with Action MLP output

The attached architecture notes explicitly distinguish flow-matching from autoregressive VLA behavior through action conditioning, attention masking, prediction objective, and generation strategy. filecite turn27file0

3. QVLA Model Architecture & Component Map

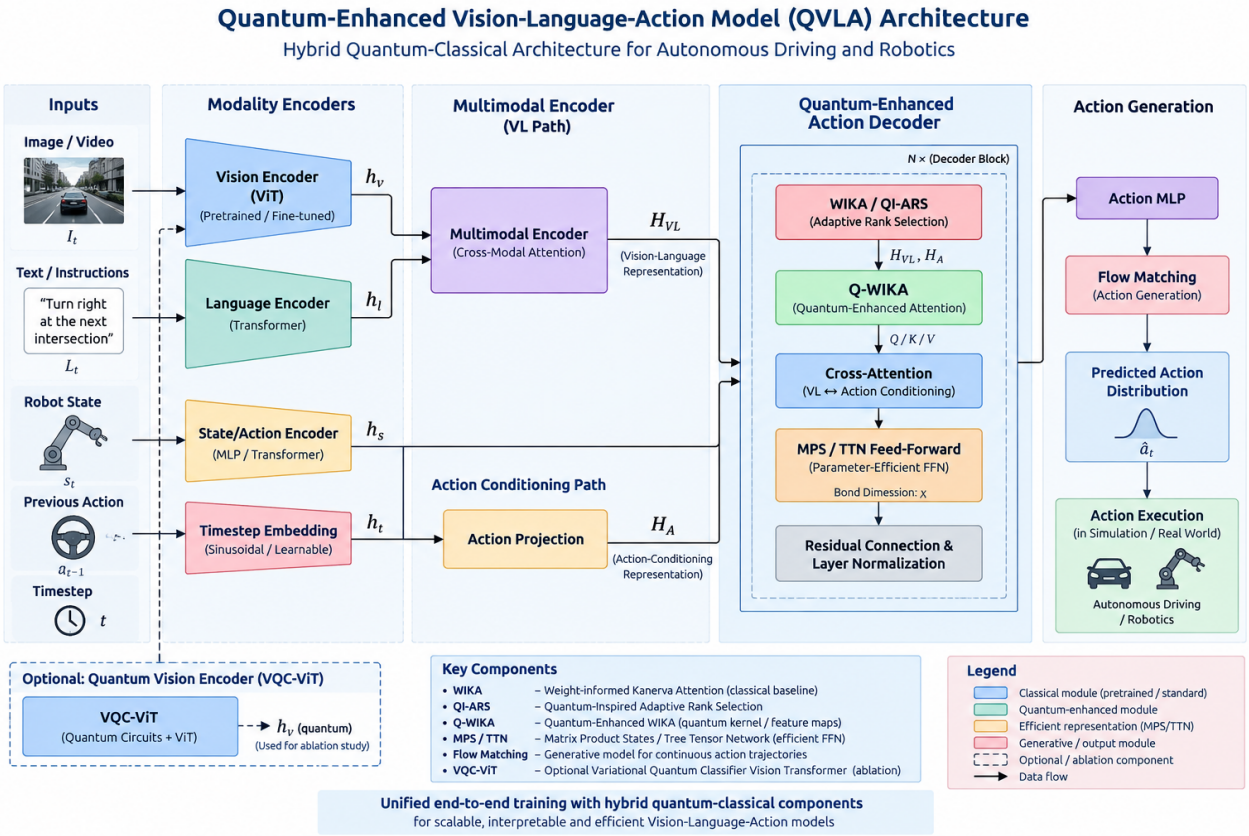


Figure S2. Detailed model-architecture infographic supplied for supplementary technical context. It complements, but does not replace, the locked canonical architecture in Figure S1.

Component-to-Bottleneck Mapping

Component	Primary purpose	What is experimentally tested
WIKA / compact WIKA	Quantum-inspired attention representation and reduced interaction cost	Representation, kernel, rank and efficiency–quality trade-offs
QI-ARS	Adaptive selection of compact attention rank	Fixed-r vs. spectral/effective-rank vs. task-aware rank
Q-WIKA	Genuine quantum feature-map extension of WIKA	Simulator → noise-aware simulation → QPU; quantum overhead and value
MPS / TTN	Tensor-network parameterization of selected decoder matrices	Bond-dimension χ sweep, fine-tuning, footprint and quality
Flow Matching	Continuous multi-action generation	Action quality, trajectory quality, latency and end-to-end behavior

Supplementary interpretation: The detailed infographic should be read together with the locked canonical diagram and the main proposal. The canonical architecture remains authoritative; the supplementary visual provides a richer component map for reviewers who want additional technical detail.

Recommended reading order: Main 6-page proposal → Figure S1 for the authoritative architecture → classical VLA reference table → Figure S2 for detailed component context.